Strong generalization for natural deduction

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openlogic:first-order-logic/natural-deduction:local-snapshot-20261013; CC BY 4.0

Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel thm:strong-generalization of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the universal-introduction rule of natural deduction and its eigenvariable condition
- derivations from open assumptions

Theorem 0.1. *If c is a constant not occurring in Γ or $!A(x)$ and $\Gamma \vdash !A(c)$, then $\Gamma \vdash \forall[x][!A(x)]$.*

Proof. Let δ be !!aderivation of $!A(c)$ from Γ. By adding a inference, we obtain !!aderivation of $\forall[x][!A(x)]$. Since c does not occur in Γ or $!A(x)$, the eigenvariable condition is satisfied. □

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